

Bradykinin-Induced Pulmonary Angioedema: A Curable Form of ARDS in COVID-19 Patients?

Thomas T. Ma^{1*}, Xue-Song Sun¹, Yu-Jia Zhai¹, Douglas C. Eaton¹

¹Emory University, Atlanta, Georgia, USA

Email: thomas.ma@emory.edu, +1 (678) 672-7957

Abstract

Acute respiratory distress syndrome (ARDS) is the leading cause of death in COVID-19. Gene expression analysis of bronchoalveolar cells reveals that SARS-CoV-2 infection induces a severe bradykinin storm, suggesting a possible link between bradykinin-mediated vascular permeability and pulmonary angioedema. We hypothesize that bradykinin-induced pulmonary angioedema represents an early, potentially reversible stage of ARDS that can be blocked by the bradykinin B₂ receptor antagonist icatibant. Using intravital multiphoton microscopy in live mice, we found that intratracheal bradykinin administration causes lung capillaries to leak large molecules (albumin and dextran), reduces alveolar airspace, and increases lung wet weight—consistent with pulmonary angioedema.

Because the renin–angiotensin system interacts with bradykinin signaling via the AT₁R–B₂R heterodimer, we investigated whether angiotensin II (Ang II) contributes to endothelial injury. Preliminary data show that Ang II induces lung endothelial cell death through the AT₁R–B₂R heterodimer, and that this effect is blocked by [Sar¹,Ile⁴,Ile⁸]-Ang II (SII-Ang II). Because this heterodimer is mechanosensitive, it responds to increased mechanical stress from alveolar edema by the Ang II receptor antagonist. We propose that bradykinin-induced edema amplifies Ang II-mediated endothelial injury, leading to fatal ARDS.

Future studies will determine whether intratracheal icatibant delivery or SII-Ang II treatment can prevent bradykinin- and Ang II-mediated ARDS, providing in vivo data to support ongoing and future clinical trials targeting B₂R and AT₁R–B₂R signaling in COVID-19 patients.

Biography

Thomas Ma is pursuing a B.S. in Computer Science at Emory University with research interests at the intersection of artificial intelligence and biomedical science. His current work applies deep learning techniques to analyze flow cytometry data for multiple myeloma detection, and a graph neural network architecture for breast cancer recurrence prediction. He aims to integrate computational modeling with experimental research to advance precision diagnostics and therapeutics.